Felix Koehler

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Research scientist at a biotech startup with a PhD from TUM specializing in deep learning for physical simulations, with publications at NeurIPS and ICLR. Expertise in adjoint methods, autodiff, numerical simulators, fluid/solid mechanics, inverse problems, optimization theory, and neural emulators. Creator of the largest SciML YouTube channel with >34k subscribers.

Experience

VIRDX
Research Scientist — since September 2026
- Biophysical simulations and deep learning systems for non-invasive tissue characterization using medical imaging.
- VIRDX is part of QuantCo.

Meta
Research Scientist Intern — June 2025 - October 2025
supervised by Ryan Goldade
- Developed Graph Neural Network emulators to accelerate 3D softbody physics-based simulation for contact and collision.

Siemens
Research Student — March 2020 - April 2022
supervised by Dirk Hartmann
- Researched explainable & efficient explicit approximations to model predictive control strategies with patent filed (US20240176310).

Volkswagen
Research Intern — October 2018 - March 2019
- Enhanced topology optimization with adjoint-based sensitivities for up to 40% weight reduction; results presented at 13th WCSMO (Paper).

Publications

Compute-optimal data scaling for neural surrogates via multi-fidelity training. (ArXiv, Project Page) — NeurIPS 2026
Neural Emulator Superiority: When Machine Learning for PDEs Surpasses its Training Data. (ArXiv, Project Page) — NeurIPS 2025
PRDP: Progressively Refined Differentiable Physics. (Shared First-Authorship). (ArXiv, Project Page) — ICLR 2025
APEBench: A Benchmark for Autoregressive Neural Emulators of PDEs. (ArXiv, Code, Project Page) — NeurIPS 2024

Education

PhD Computer Science (Dr. rer. nat.)
Technical University of Munich — September 2022 - August 2026
supervised by Nils Thuerey
- Coordinated annual lecture in *GamePhysics* (applied numerical methods + mechanical modeling) for >200 students.
- Supervised multiple students, e.g., Unrolled vs. Implicit Autodiff for Linear System Solves by Kanishk Bhatia.
- Instructed a lecture on *Autodiff and Adjoints in Differentiable Physics* for our *Physics-Based DL* master course (Recording, Slides).
- Orchestraded large-scale GPU training runs on LRZ & Leonardo supercomputers.

M.Sc. Computational Science & Engineering (CSE)
Technical University of Munich — October 2019 - April 2022
GPA: 1.2 *(German system: 1.0 (best) to 6.0 (worst))*, best 2%
- Relevant course work (fully in English):
  - Numerical Analysis, Scientific Computing, Nonlinear Finite Element Method, Computational Plasticity.
  - Machine Learning, ML for Graphs & Sequential Data, Uncertainty Quantification, Probabilistic Machine Learning, Visualization.
  - Parallel Programming, High-Performance Computing, Parallel Numerics.
- Honors: Scholarship by *Studienstiftung des Deutschen Volkes* (most prestigious national scholarship program in Germany).
- Exchange semester at *KTH Stockholm* (August 2020 - January 2021).
- Activities: Electrodynamical Simulations of magnetic lift and drag with *ANSYS Maxwell* for TUMHyperloop.

B.Sc. Mechanical Engineering
Technical University of Braunschweig — October 2015 - January 2019
GPA: 1.1 *(German system: 1.0 (best) to 6.0 (worst))*

Skills

- Python: JAX, Tensorflow 2, PyTorch (Lightning), Matplotlib, Pandas, Seaborn, Scikit-Learn, SciPy, NumPy, FEniCs, WandB
- Other programing languages: C++ (OpenMP, MPI, Eigen, CMake), Julia, bash, FORTRAN, MATLAB
- Tools/Platforms: Linux (Ubuntu & Arch), Git, Slurm, Docker, AWS/GCP/Azure, OpenFoam, Paraview, ANSYS Maxwell, Siemens Amesim
- Spoken languages: English (professional, IELTS 8.5/9.0), German (native), French & Swedish (elementary)

Projects & Open Source

- Founder and producer of the Machine Learning & Simulation YouTube channel; created more than 200 videos on AI for Science and adjacent topics (autodiff, optimization, probability, neural networks etc.), e.g., FNO, DeepONet, Adjoint ODEs, Variational Inference.
- Exponax: I designed and maintain a JAX-based Fourier-spectral PDE solvers library based on ETDRK methods (Docs).
- Co-Organizer of *Differentiable Systems and SciML Workshop* at EurIPS 2025.
- Gold Reviewer for ICML 2026, Top Reviewer for NeurIPS 2026.